



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SUBRAMANIAM PETROLEUMS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

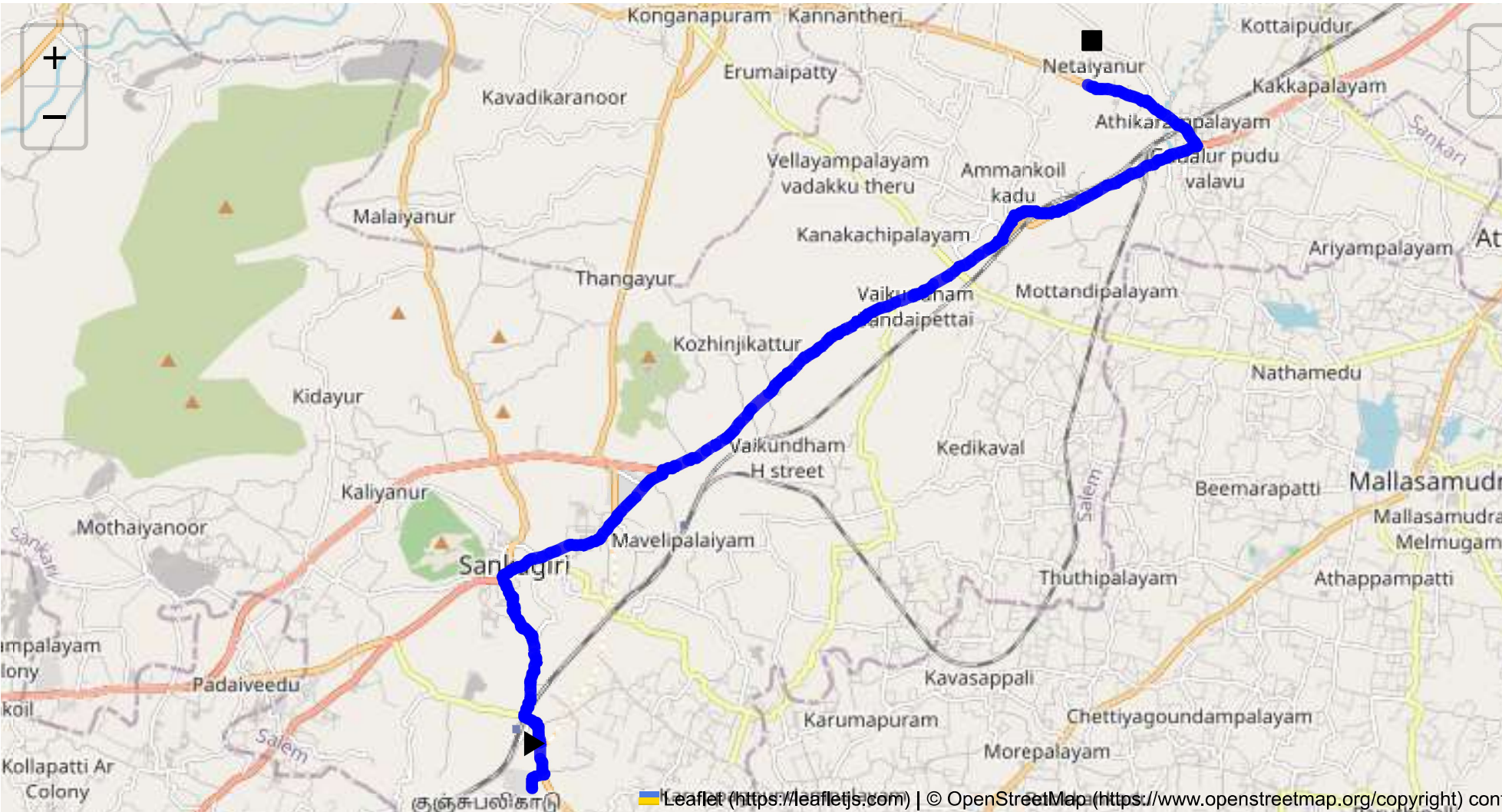
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 22.85 km
Estimated Duration: 0.5 hours
Adjusted Duration (Heavy Vehicle): 0.7 hours
Start: (11.4381, 77.8734)
End: (11.557896, 77.97023)



Welcome to the Journey Risk Management Study

To perform a comprehensive analysis of the route from CVQF+23W, Sangagiri to Petrol Bunk, Idappadi Rd, Eranaapuram, here's a breakdown based on the points you've provided:

1. Overview of the Route Map:

- The route is approximately 22.85 kilometers long, predominantly following local roads through rural and semi-urban areas in Tamil Nadu.

- The journey typically includes navigating a series of village roads, state highways, and may require passing through small towns.

2. **Typical Weather Conditions and Potential Weather-Related Hazards:**

- The climate around this region is generally tropical, with hot summers and moderate monsoons.
- During the monsoon (June to September), roads can become slippery and visibility may decrease due to heavy rains, increasing the risk of hydroplaning.
- Ensure to monitor weather forecasts to avoid driving through heavy rainfall which can lead to road inundation.

3. **Traffic Patterns, Peak Hours, and Congestion-Prone Areas:**

- Traffic congestion is less typical for rural roads but expect slow-moving traffic within towns and near intersections, especially during morning (8-10 AM) and evening (5-7 PM) hours.
- Be cautious near marketplaces and schools where pedestrian activity can peak.

4. **Assessment of Road Quality and Infrastructure:**

- The roads are mostly single-lane and may have uneven surfaces, though state highways are generally in better condition.
- Be watchful for potholes and unmarked speed breakers, especially after the rainy season when road deterioration is common.

5. **Suggestions for Alternative Routes for Emergencies:**

- If the main route is blocked, consider taking alternative state highways running parallel but ensure they accommodate the size and restrictions of your vehicle.
- Local maps and GPS systems will be helpful in identifying feasible alternative roads when needed.

6. **Summary of Local Regulations Affecting Hazardous Material Transport:**

- Ensure to adhere to local regulations regarding the transport of hazardous materials, which may include specific timings when transport is allowed and restrictions through certain residential areas or during school hours.
- Proper signage on vehicles and possession of required permits is mandatory.

7. **Historical Incidents Involving Heavy Vehicles or Hazardous Materials:**

- Records may show sporadic incidents due to overloading and speeding or during inclement weather conditions.
- There have been isolated incidents relating to overturning on sharp bends or small bridges due to speed.

8. **Environmental Considerations and Sensitive Areas:**

- Be cautious passing through agricultural zones to avoid any spillage that could affect crops.
- Maintain low noise levels in the vicinity of schools and hospitals marked by road signs.

9. **Analysis of Communication Coverage, Noting Potential Dead Zones:**

- While major towns have good communication networks, some stretches between villages might have weakened mobile coverage.

- Reliance on physical maps and having offline navigation options could be prudent here.

10. Estimated Emergency Response Times for Different Route Segments:

- Rural segments might have delayed emergency response times due to distance from emergency services.
- Best practice involves being prepared with first aid and having contact details of local emergency services on hand.

11. Overall Summary of Risk Assessment:

- The overall risk for the route is moderate with increased risk during adverse weather conditions.
- Proactive measures include pre-trip inspection, defensive driving, and real-time weather monitoring to mitigate risks effectively.
- Emphasis should be on adherence to road safety regulations and continuous assessment of road conditions during transit.

In conclusion, while the route is generally safe for travel, caution is advised regarding weather conditions and navigating through smaller roads/areas with higher pedestrian activity. Regular updates on local road conditions and emergency plans will further enhance safety for truck drivers carrying hazardous materials on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	High	11.48369, 77.88781	15 KM/Hr
7	Turn	Medium	11.48370, 77.88787	30 KM/Hr
8	Turn	Medium	11.49149, 77.89601	30 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Blind Spot	Blind Spot	11.54736, 77.98881	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47407, 77.86839



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48369, 77.88781



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48370, 77.88787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.54736, 77.98881

